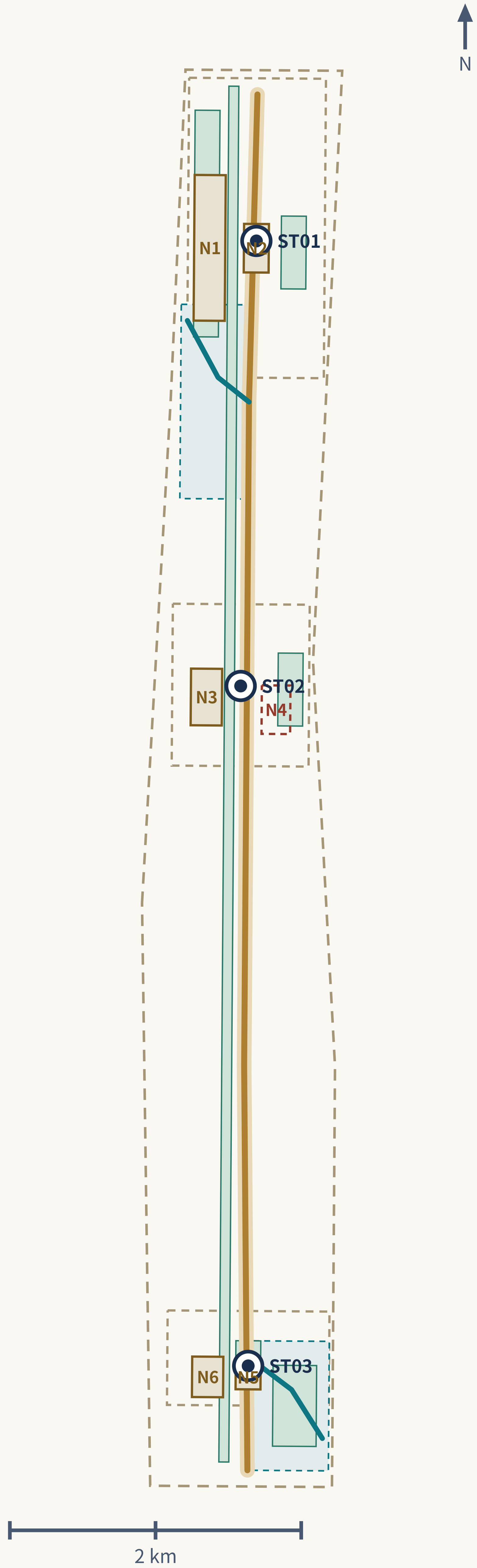


One spine, three stations, two wings



ST01 • Exchange and dispatch

Zhongzhiyuan AI Independent Innovation Acceleration Area

Compute Exchange

Positioning

Full-stack independent innovation and AI-governance capability; the new axis’ s place for deciding and explaining where tasks are computed.

Spatial structure

A Qinghe-front public living room organized as river → garden → courtyard → R&D cluster; dispatch and testing gather in public-facing central courtyards.

AI scenarios

02 Safety Governance Sandbox, 06 Jing-Zhang Heritage and AI Public Education, 08 Haidian–Ulanqab Green Compute Dispatch Test, 10 Climate Digital-Twin Operations Test.

Concept nodes

N1 Qinghe Compute Commons — Waterfront linear park + a group of open courtyards
N2 Fail-safe Test Yard — Hard-surfaced yard that can be closed or opened + outer gallery

ST02 • Verification and incubation

Beijing AI Origin Community

Origin Commons

Positioning

Validation and incubation for a world-class AI ecosystem; public services receive community and university review before citywide use.

Spatial structure

Courtyard-centered small-scale network linking campus, park, and neighborhood; open-source honors line the slow-mobility spine.

AI scenarios

01 Open-Source Launch Hall, 04 Newcomer 72-Hour Settling-In, 06 Jing-Zhang Heritage and AI Public Education, 09 Research-to-SME Conversion Test.

Concept nodes

N3 Origin Courtyard — Semi-enclosed community courtyard + open street-level ground floors
N4 Campus–Station Stitch — A continuous walking-and-cycling stitch with option comparison

ST03 • Launch and translation

Dazhongsi AI Industry Cluster

Axis Launch Station

Positioning

Launch and conversion area for intelligence-native business; the axis’ s public, market, and international interface.

Spatial structure

Market-led, with four-quadrant pedestrian continuity and ground-floor services around rail stations; launch and cultural content follow the walking network.

AI scenarios

05 Barrier-Free Continuous Passage, 07 Safe Route Home for Night Workers, 03 Network/Power Fail-safe, 10 Climate Digital-Twin Operations Test.

Concept nodes

N5 Four-Quadrant Crossing — Continuous walking surfaces and waiting spaces at four corners
N6 New Axis Launch Market — Adaptable open street frontage + a permanent canopy

Six urban spaces: a repeatable, combinable block grammar

Space type	Urban role	Typical design action	Layer
Gateway	Information, accessibility and public-service interface	Station entrances, wayfinding, accessible ramps and a staffed desk in authorized windows	N5
Courtyard	Exchange, community courses, rest and small trials	Semi-open courtyard, bookable test bays, shade and seating	LU-004
Station	Exchange node for compute, mobility, service or events	Edge compute, transfer and event gathering with parallel digital and non-digital entrances	ROAD-AXIS-001
Market	Showcase, procurement, culture and everyday commerce	Open street frontage, activated ground floor, temporary exhibits	LU-003
Garden	Shade, winter sun, stormwater detention and quiet rest	Climate public space, floodable green, rain gardens	GREEN-AXIS-001
Stitch	East-west links across rail, road and water barriers	Active-travel stitching and ecological links; feasibility studied separately	ROAD-STITCH-N4

The six spaces are combinable prototypes for the professional teams that follow. The spine begins as free, continuous, accessible and climate-adapted public space, then layers wayfinding, service and smart facilities to strengthen movement, shade, rest and accessibility.

Two wings and the coordination loop

Zhongguancun technology-service wing

Factor supply: capital, legal, IP, talent and international networks

Xiaoyue River scenario-enablement wing

Real-demand feedback: stress-testing community, ecology, mobility and everyday needs

Loop direction: Origin Community verifies → Zhongzhiyuan exchanges and dispatches → Dazhongsi launches and translates → the two wings supply factors and real demand → back to Origin Community for the next round.

Three positionings

Centennial Jing-Zhang cultural belt, urban AI everyday-life belt, AI convergence innovation belt.

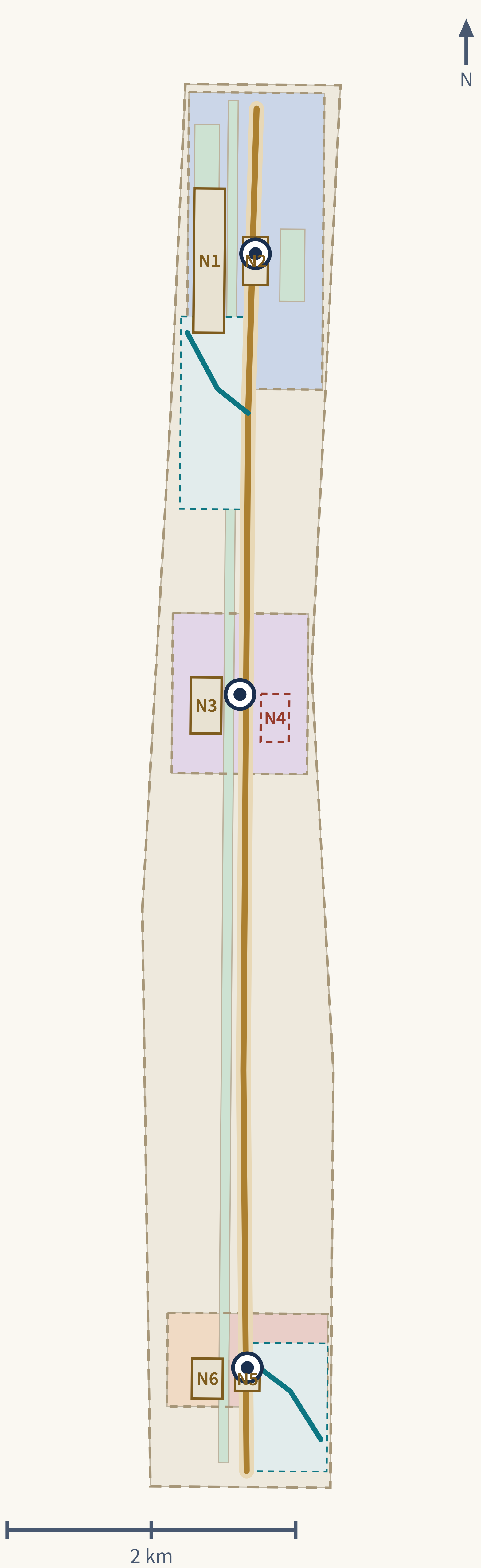
Five functions

A full-stack independent AI innovation system, a world-class AI innovation ecosystem, a new AI+ scenario-enablement paradigm, an intelligent and lively AI city, and a global voice in AI governance.

Localized mechanism chain

Climate public space → research and talent anchor → a standing operating body → twin-city green compute cooperation → a digital twin with a rights boundary. The order is itself the design judgement: make the public space good without any device first, then layer the smart systems on.

Conceptual land use and renewal landing points



Conceptual land-use partition (closed, gapless, non-overlapping; classification follows the national land-use guide)

Feature	Name	Code	Area	Derivation
LU-001	AI R&D innovation land (diagrammatic concept)	0802	144.2 ha	PROV-KEY-001 minus the union of proposed green-space concept polygons that overlap it, clipped to SITE-001; computed and recomputed in EPSG:4548 with a shared 1mm snappin
LU-002	Community service facility land (diagrammatic concept)	0702	88.2 ha	PROV-KEY-002 minus the union of proposed green-space concept polygons that overlap it, clipped to SITE-001; computed and recomputed in EPSG:4548 with a shared 1mm snappin
LU-003	Commercial and enterprise service land (diagrammatic concept)	05	23.3 ha	PROV-KEY-003 minus the union of proposed green-space concept polygons, split at longitude 116.347 so the market-node (N6) side is retained; clipped to SITE-001; computed
LU-004	Cultural / railway heritage interpretation land (diagrammatic concept)	0803	31.2 ha	PROV-KEY-003 minus the union of proposed green-space concept polygons, split at longitude 116.347 so the heritage/crossing-node (N5) side is retained; clipped to SITE-001
LU-005	Park and open space land (diagrammatic concept)	1401	129.5 ha	Union of all six submitted green-space concept polygons (GREEN-AXIS-001, GREEN-QINGHE-001, GREEN-XIAOYUE-001, GREEN-ST01-001, GREEN-ST02-001, GREEN-ST03-001), clipped to
LU-006	Reversible reserve / flexible land (diagrammatic concept)	16	724.9 ha	SITE-001 minus the union of LU-001..LU-005; computed and recomputed in EPSG:4548 with a shared 1mm snapping grid so shared boundaries with adjacent land-use polygons use

Renewal portfolio for professional development

ID	Project	Key dependencies	Deepening focus
HX-01	Spine walking/cycling gap stitching and continuous shade	Road right-of-way, space under bridges, traffic review	Use gap surveys and traffic simulation to select ring-road crossing solutions
HX-02	Qinghe–Xiaoyue blue-green cross-link and detention	River blue line, ecological and flood conditions	Calibrate detention capacity with hydrology and engineering data
HX-03	Origin Commons and open-source honour display	Campus boundary, ownership, ground-floor uses	Use a co-creation charter to link campus, firms and communities
HX-04	Compute Exchange and public ledger interface	Energy, safety, operator and the Haidian–Ulanqab agreement	Use the project agreement to integrate real ledger data
HX-05	Dazhongsi four-quadrant pedestrian connection	Rail station, intersection, utilities	Select the engineering scheme through multi-agency option comparison
HX-06	Spine accessibility continuity and night-safety upgrade	Condition survey, lighting and maintenance body	Use existing-condition surveys to set continuity and night-service baselines
HX-07	Axis Timetable and annual open test season	Public-space permits, event safety, rights clearance	Let the Open Operations Consortium establish an annual budget and durable program

Three reversible phases and the four-tier building strategy

90-day standards and no-personal-data drill (concept)

90-day standards rehearsal: establish rules, definitions, process and ledger structure with public, synthetic or de-identified data.

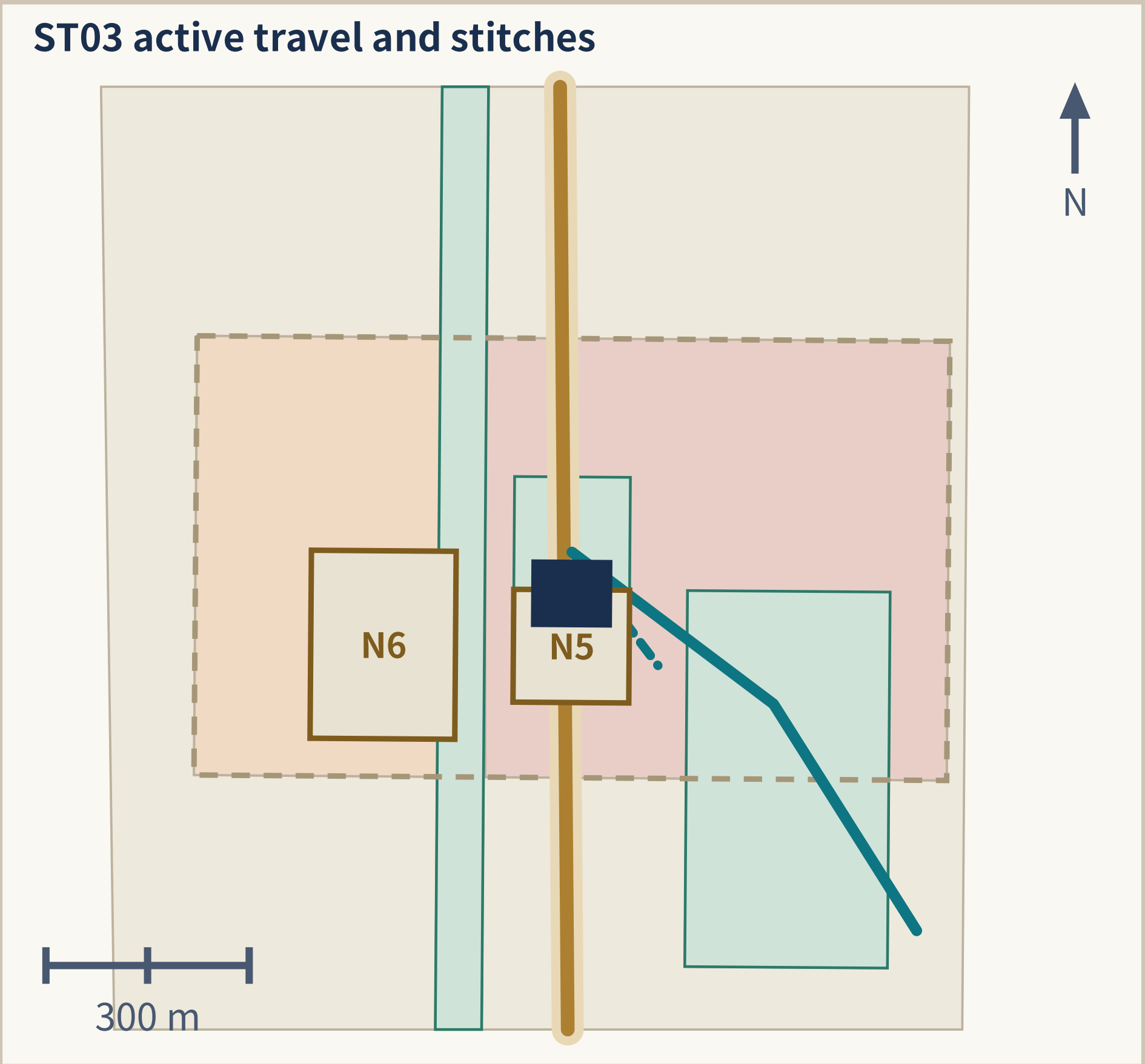
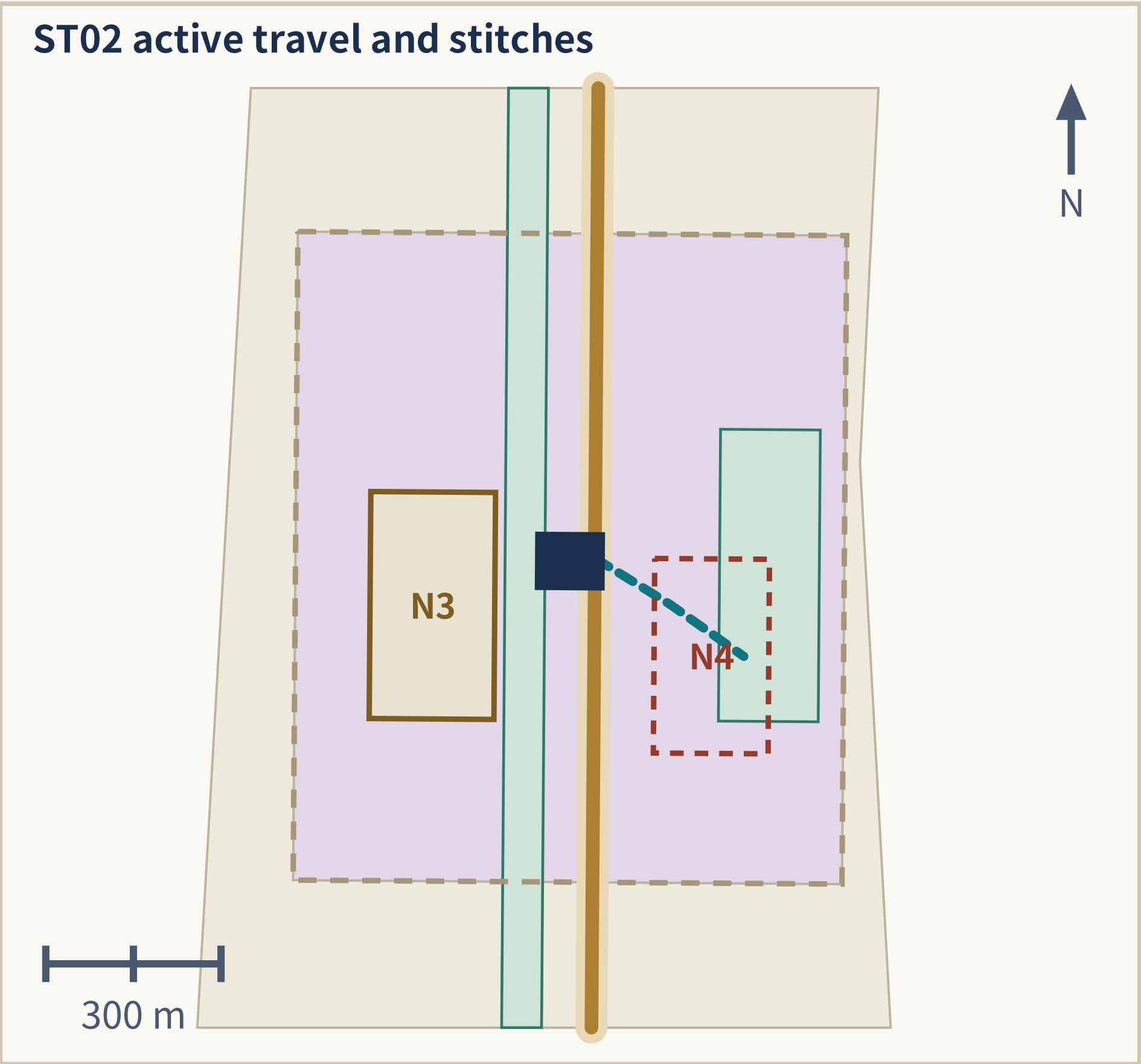
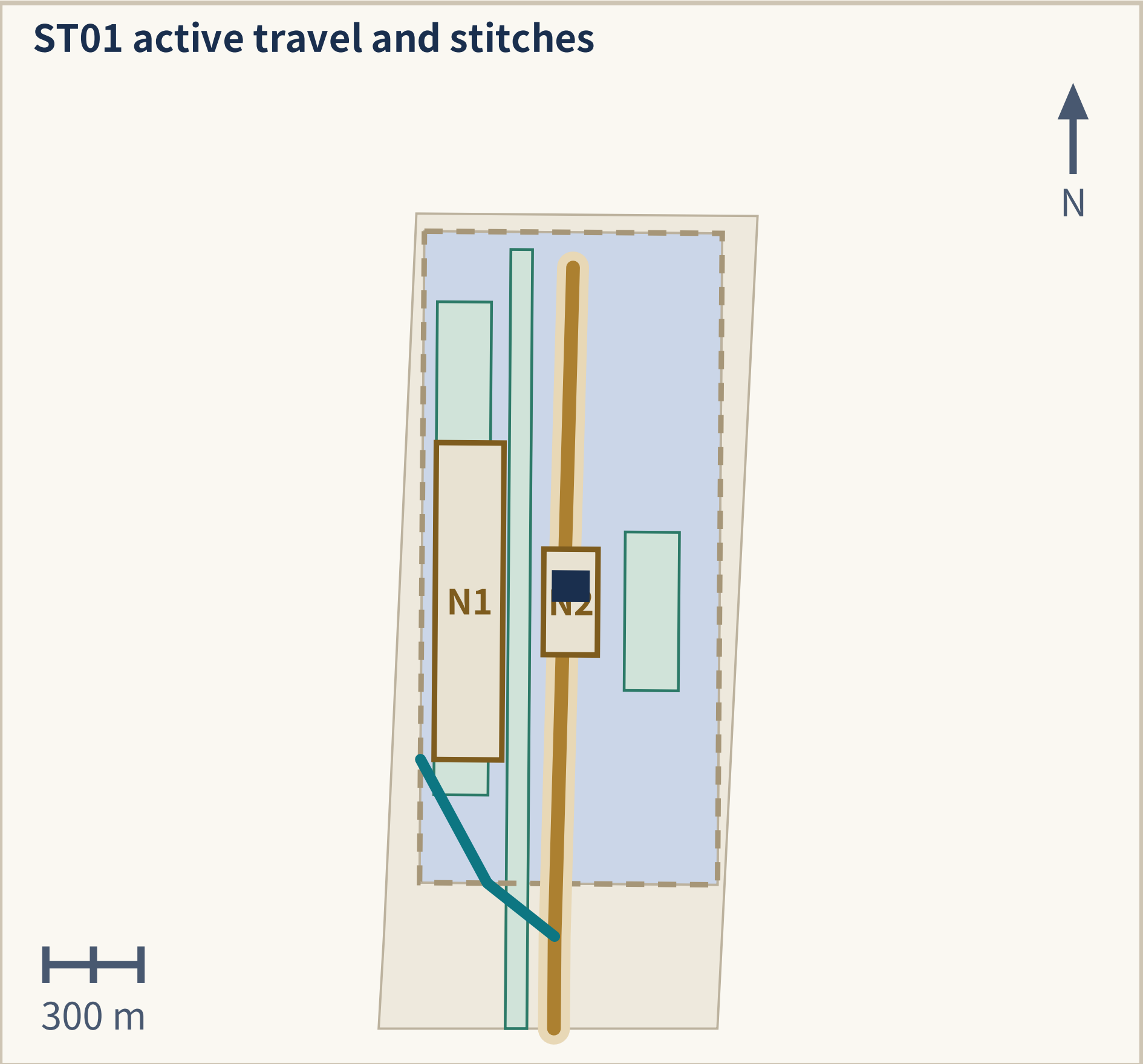
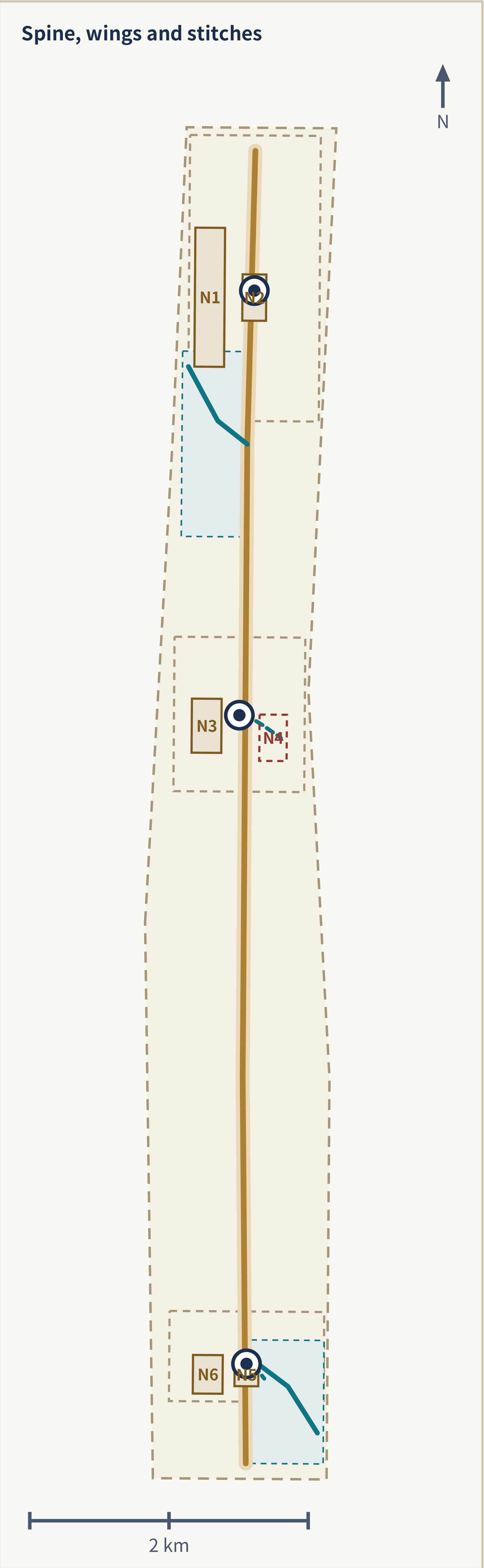
One-year reversible public pilot (concept)

One-year reversible public pilot: run at selected nodes and continuously adjust facilities, rules and service from evidence.

Annual renew, modify, or stop cycle (concept)

Annual audit and renew / modify / stop decision: independent audit supports transparent evidence-led decisions.

Tier	Design action
Retain	Keep massing and frontage; maintenance only
Light renovation	Activate ground floor, open frontage, tidy roof, add accessibility
Reversible addition	Removable, relocatable lightweight structures
New build pending verification	Position intent only; no scale or height



Five conceptual alignments (lengths recomputed in EPSG:4548; not engineering alignments)

Feature	Name	Class	Nodes	Length
ROAD-AXIS-001	North-south public greenway spine (concept)	greenway	N1 N2 N3 N4 N5 N6	9,439 m
ROAD-WING-TECH-001	Zhongguancun technology-service wing link (concept)	pedestrian	N1 N2	714 m
ROAD-WING-XIAOYUE-001	Xiaoyue River scenario-enablement wing link (concept)	greenway	N5 N6	768 m
ROAD-STITCH-N4	Campus-Station Stitch connector (concept)	transit_connection	N4	306 m
ROAD-STITCH-N5	Four-Quadrant Crossing stitch (concept)	pedestrian	N5	210 m
Total	conceptual_mobility_length_m			11,437.434 m

N4 Campus–Station Stitch research task

N4 Campus–Station Stitch is a dedicated research task. Transport, accessibility and engineering studies will compare crossing, end-space and operating options to form a safe, continuous and legible public link between campus and station.

Technical record: design_only_closed_pending_study; the opening scheme follows the completed study.

Utilities and new infrastructure: turning invisible compute into sited, accountable urban facilities

Edge compute points

Placed in station-type spaces for low-latency public services; always co-located with rest, enquiry and charging, never as standalone machine rooms.

Climate nodes

Combined with gardens and courtyards for shade, warmth, drinking water and temporary storm refuge; the physical carrier of the extreme-weather modes.

City-side expression of failover

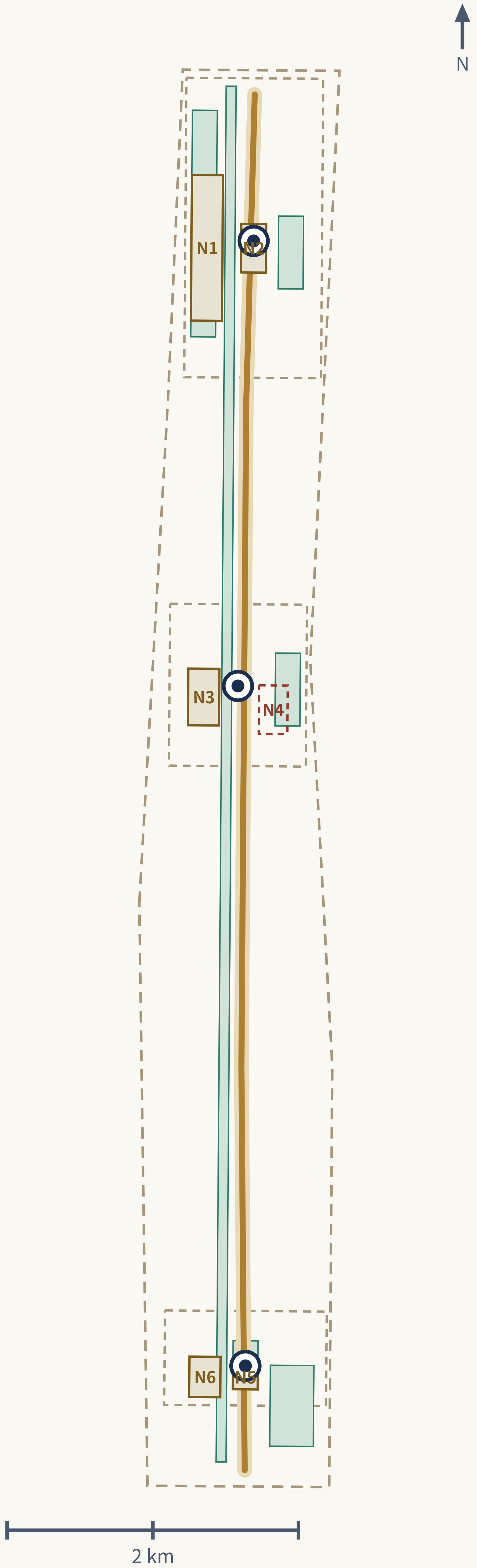
When an external link fails, stations display a degraded state and switch to staffed service; a black screen with nobody to ask is not acceptable.

Conventional utilities

Existing water, drainage, power, telecom and sanitation capacity is unknown and is listed as a precondition for formal development; no load calculation is offered.

The CONSTRAINTS surface is the professional coordination interface for the next planning, mobility, utility, ecology and heritage inputs.

Blue-green network and heritage spine



Six blue-green concept surfaces (recomputed areas)

Feature	Name	Area
GREEN-AXIS-001	Heritage spine continuous shade corridor (concept)	64.5 ha
GREEN-QINGHE-001	Qinghe riverside detention green belt (concept)	26.5 ha
GREEN-XIAOYUE-001	Xiaoyue River seasonal refuge green space (concept)	16.6 ha
GREEN-ST01-001	Zhongzhiyuan station commons green (concept)	8.5 ha
GREEN-ST02-001	AI Origin Community station commons green (concept)	8.5 ha
GREEN-ST03-001	Dazhongsi station commons green (concept)	4.7 ha
Union	green_ratio = 0.113434	129.5 ha

Six public-space prototype nodes (recomputed areas)

Node	Name	Prototype	Area
N1	Qinghe Compute Commons	Garden + Courtyard	21.3 ha
N2	Fail-safe Test Yard	Courtyard + Station	5.7 ha
N3	Origin Courtyard	Courtyard + Gateway	8.3 ha
N4	Campus–Station Stitch	Stitch + Station (research task)	6.5 ha
N5	Four-Quadrant Crossing	Gateway + Station	2.8 ha
N6	New Axis Launch Market	Market + Courtyard	5.9 ha
Union	public_space_ratio = 0.044371		50.6 ha

A climate-first public spine: everyday activity, storm detention, safe recovery

Everyday mode

Continuous shade, ventilation, seating and rest/cooling points; establish coverage through walking-distance control.

Storm-detention mode

Activate tiered storage, floodable green and rain gardens; set capacity through hydrology and engineering data.

Safe-recovery mode

Run desilting, inspection and reopening; human collaboration is a standard resilience process.

Three AI pilgrimage landmarks and cultural narrative (concept)

Compute Exchange (Zhongzhiyuan)

Centred on a publicly readable dispatch and ledger interface that turns “where a task is computed” into visitable, questionable public content; the honour display shows public test records and evaluation reports, not company rankings.

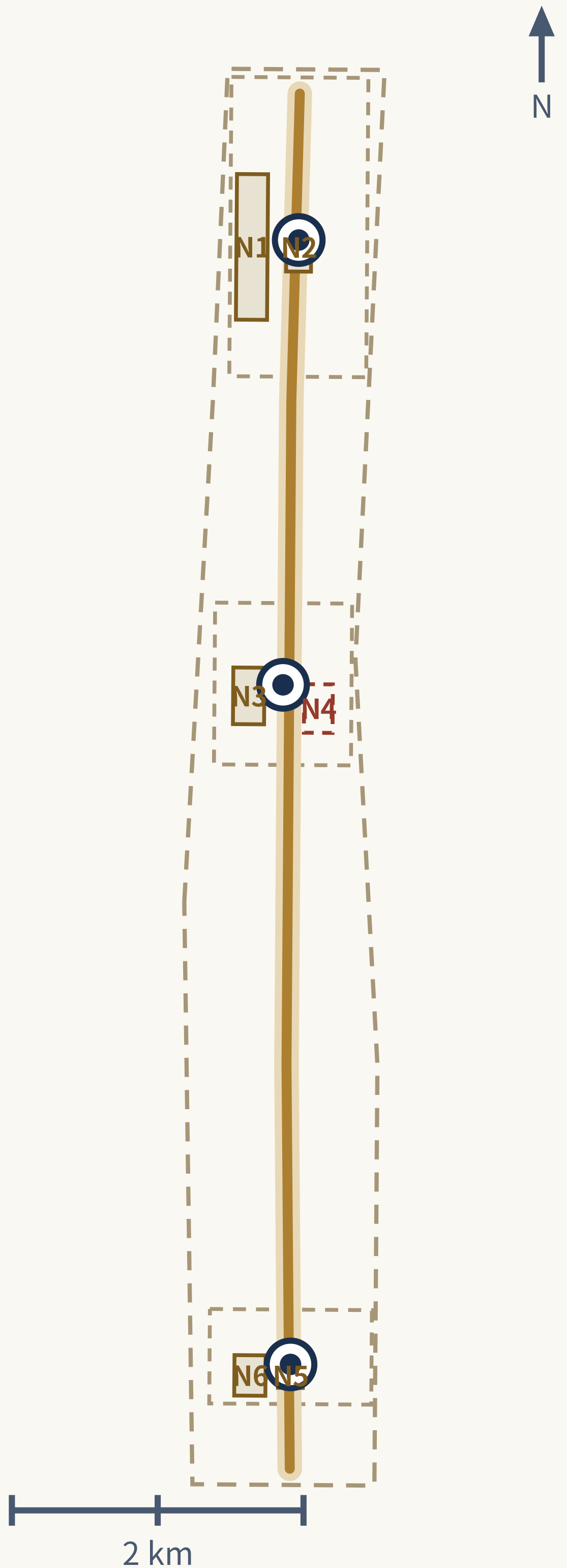
Origin Commons (AI Origin Community)

Centred on an open-source honour wall, community oral history and live model verification, recording contributors and contribution events; any display can be removed at a contributor's request.

Axis Launch Station (Dazhongsi)

Centred on the annual Axis Timetable and result launches while doubling as an everyday urban living room; international exchange and local everyday use share the same space.

Index



N4 stitch study

N4 Campus–Station Stitch is a dedicated research task. Transport, accessibility and engineering studies will compare crossing, end-space and operating options to form a safe, continuous and legible public link between campus and station.

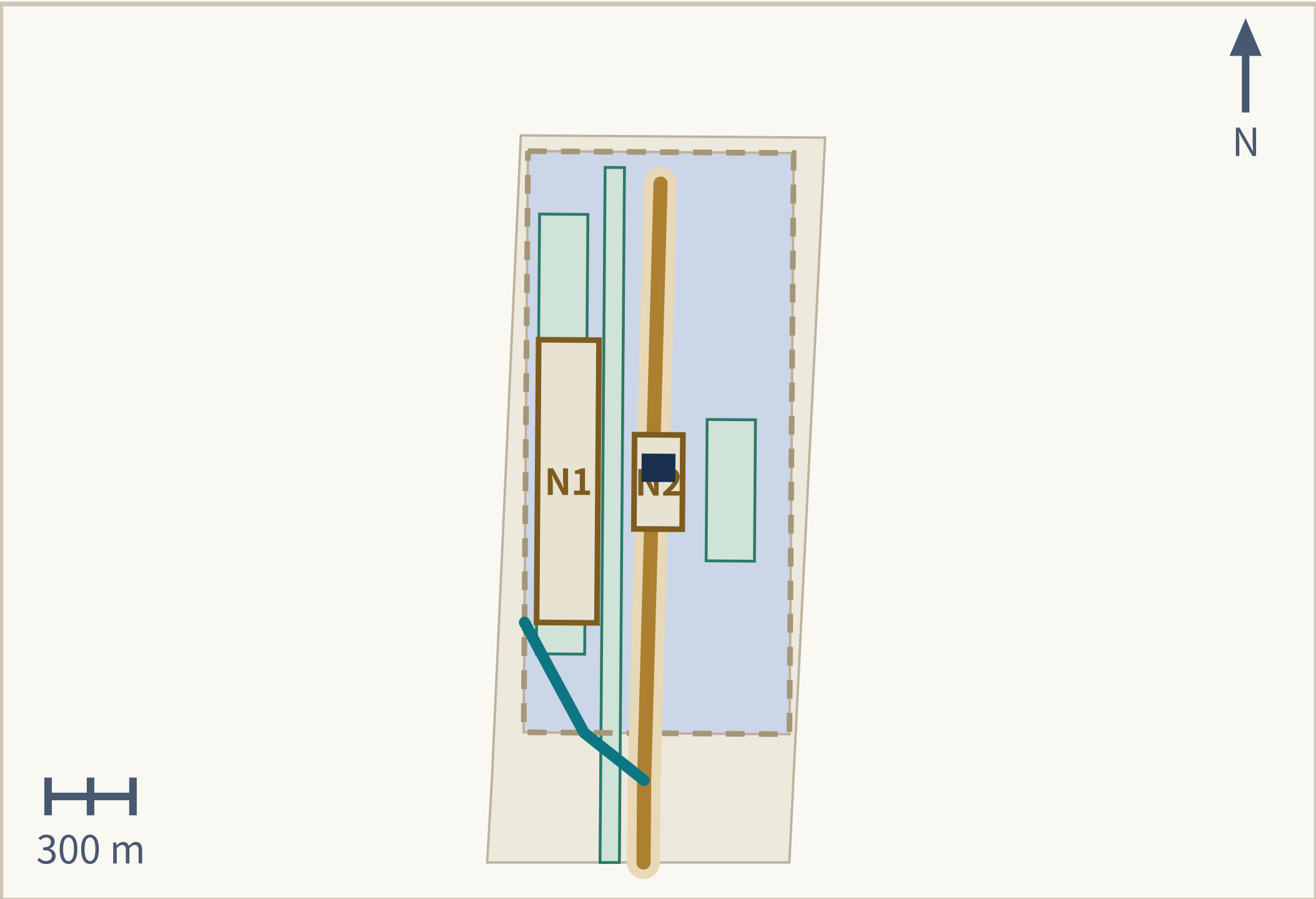
Technical record: design_only_closed_pending_study; the opening scheme follows the completed study.

ST01 • Exchange and dispatch

Zhongzhiyuan AI Independent Innovation Acceleration Area

Compute Exchange

PROV-KEY-001 • announced approx. 192.1 ha • recomputed 192.9 ha



Carries full-stack independent innovation and AI-governance voice; the only node on the axis that decides and displays where a task is computed.

N1 Qinghe Compute Commons

Waterfront linear park + a group of open courtyards

Operating modes

Daytime waterfront rest and viewing; lower night lighting with continuous wayfinding and callable human help; courtyards as shade/rain shelter and low sections as floodable green during storms.

Opportunity

Turn the Qinghe edge into a public living room where compute becomes visible, understandable, and discussable.

Manual / offline fallback (declared geometry field, quoted verbatim)

Printed/static signage is on-site; a staffed help point is a design target during authorized service periods only, subject to confirmed operator, and is not a permanent-staffing claim. The dispatch ledger interface has an offline static-display fallback and never requires a digital identity to view.

N2 Fail-safe Test Yard

Hard-surfaced yard that can be closed or opened + outer gallery

Operating modes

Public court or booked test yard by day; ordinary open space at night; shelter and gathering during extreme weather.

Opportunity

Create a public window into system testing and human takeover.

Manual / offline fallback (declared geometry field, quoted verbatim)

Human takeover is the default action for any drill failure; printed instructions remain available on site. Staffed guidance is a design target during authorized drill/service periods only, subject to confirmed operator, with no digital step required.

Preconditions

River blue line and flood-control standard; riverside land ownership and maintenance body; energy and safety requirements for the ledger interface

Deepening focus

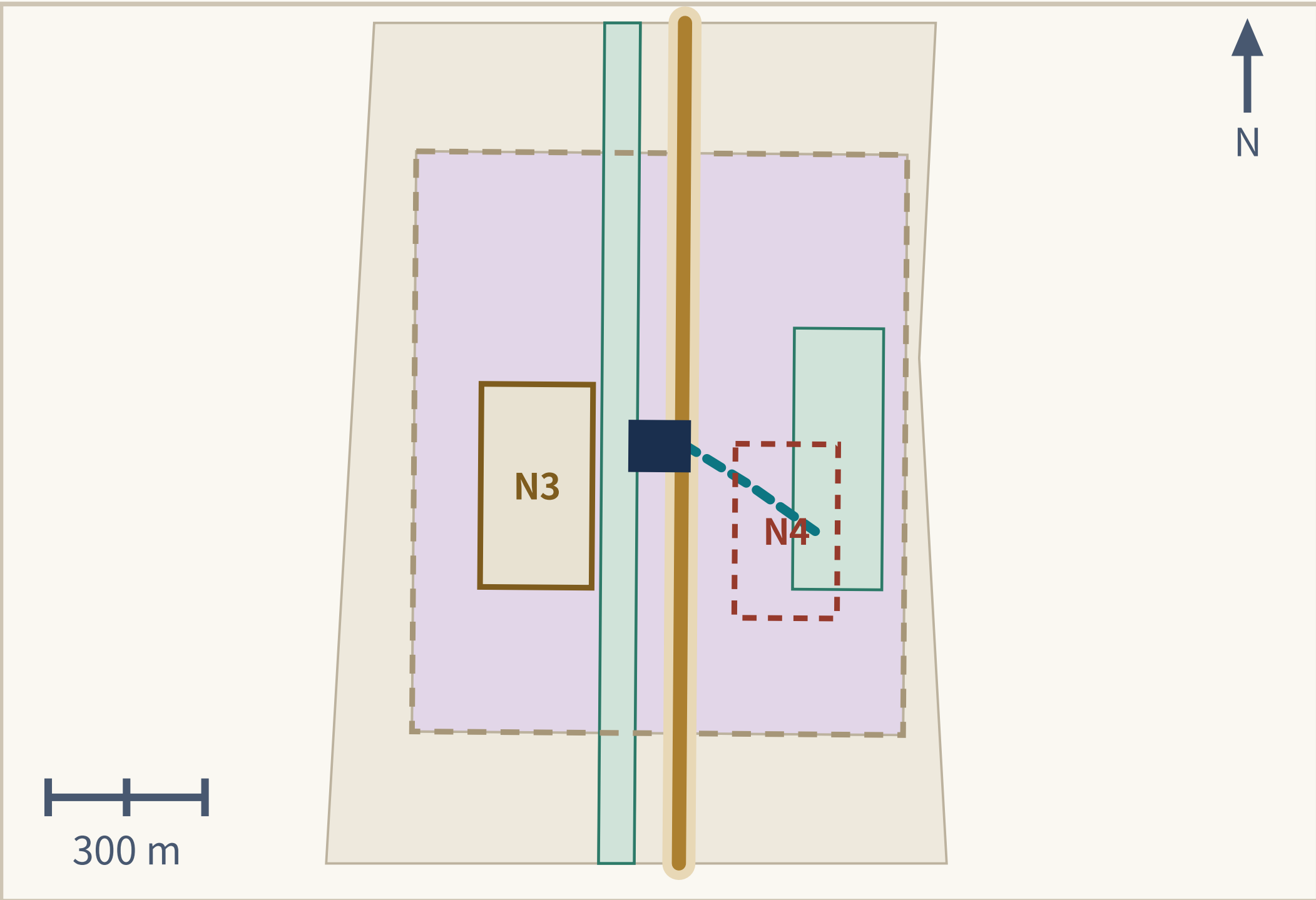
Deepening focus: coordinate river, flood, ownership, energy and safety inputs and establish time- and zone-based arrangements for public visits and professional tests.

ST02 • Verification and incubation

Beijing AI Origin Community

Origin Commons

PROV-KEY-002 • announced approx. 104.3 ha • recomputed 104.3 ha



Verification and incubation area of the world-class AI innovation ecosystem; models and public services face community and university scrutiny here before entering the city.

N3 Origin Courtyard

Semi-enclosed community courtyard + open street-level ground floors

Operating modes

Courses, validation, launches, and rest by day; lighting, seating, and callable help at night; open ground floors become cooling/warming rest points in extreme weather.

Opportunity

Give universities, companies, and residents a shared in-between place for validation, learning, and everyday exchange.

Manual / offline fallback (declared geometry field, quoted verbatim)

Printed handbook available on site; a staffed service desk is a design target during authorized service periods only, subject to confirmed operator and annual authorization, not a permanent-staffing claim. Content review and any showcase can be withdrawn without notice, no login required.

N4 Campus–Station Stitch

A continuous walking-and-cycling stitch with option comparison

Spatial intent

A continuous step-free connection with gathering space, shelter, seats, lighting, and the connector alignment ; a Stitch + Station prototype. Professional comparison selects the crossing form.

Research task (design_only_closed_pending_study)

remains closed while transport, accessibility, and engineering studies compare overpass, underpass, at-grade reconfiguration, shelter, end spaces, detours, and service arrangements.

Manual / offline fallback (declared geometry field, quoted verbatim)

The research brief includes route continuity, offline wayfinding and a safe alternative under disruption. N4 remains design_only_closed_pending_study until transport, accessibility and engineering studies confirm the feasible authorized scheme.

Preconditions

Campus boundary and access negotiation; consent for street-level ground floors; courtyard land ownership and maintenance body; resident participation and objection channels

Deepening focus

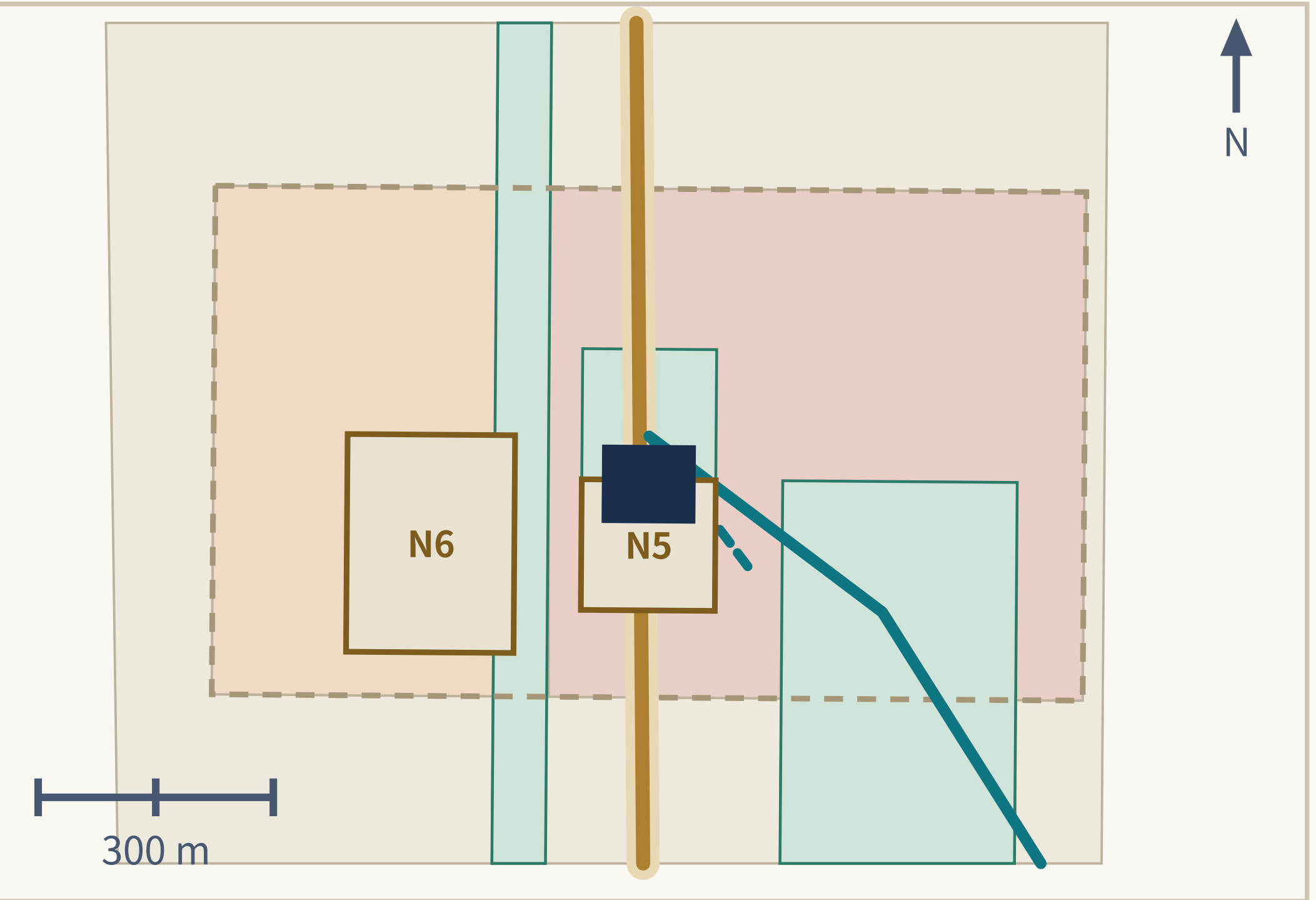
Deepening focus: establish data and research authorization, then co-design renewal and trials through building surveys and long-term community participation.

ST03 • Launch and translation

Dazhongsi AI Industry Cluster

Axis Launch Station

PROV-KEY-003 • announced approx. 72.0 ha • recomputed 72.0 ha



Launch and translation area for AI-native business forms; the axis's outward interface to market, public and international audiences.

N5 Four-Quadrant Crossing

Continuous walking surfaces and waiting spaces at four corners

Operating modes

Commuting and interchange by day; lighting, visibility, and callable help at night; shelter and guided alternate routes during rain, snow, ponding, or ice.

Opportunity

Integrate walking, transfers, waiting, and rest at all four corners of a large intersection.

Manual / offline fallback (declared geometry field, quoted verbatim)

A non-digital, continuous ground-level route with no app or digital ticket required to cross is a design target, subject to professional transport/accessibility confirmation, an authorized service window, safety conditions, and annual authorization; no existing route is claimed. Staffed guidance is a design target during authorized service periods only, subject to confirmed operator and safety conditions, not a permanent-staffing claim.

N6 New Axis Launch Market

Adaptable open street frontage + a permanent canopy

Operating modes

Market, display, meetings, and rest by day; seating and small events at night; shade and rain protection in extreme weather, with temporary elements stored for strong wind or snow.

Opportunity

Let product launch, procurement, culture, and daily street life share one year-round space.

Manual / offline fallback (declared geometry field, quoted verbatim)

Canopy and seating structure (design target) remains usable with power/network down; a printed contingency plan covers removal of temporary fixtures. Staffed presence during events is a design target subject to confirmed operator and authorized service periods, not a permanent-staffing claim.

Preconditions

Intersection safety assessment and traffic review; road right-of-way and utility conditions; rail station entrance interface; ownership split of the four corners

Deepening focus

Deepening focus: coordinate rail, road, utility and commercial partners to complete intersection, utility, ownership and public-space maintenance design.

Ten AI scenario cards sharing one rights boundary (human review, offline alternative, exit)

S01 Open-source Release Hall Origin Community	S02 Safety-governance Sandbox Zhongzhiyuan	S03 Network/power failure fail-safe Stations and gateways, whole axis	S04 Newcomer 72-hour settling Origin Community, gateways	S05 Continuous accessible movement Entire spine
S06 Jing-Zhang heritage and AI public education Heritage spine, Origin Community	S07 Safe night route for shift workers Spine and stations	S08 Haidian–Ulanqab green compute dispatch test Zhongzhiyuan AI industry test scenario	S09 Research-to-SME product translation test Origin Community, Dazhongsi AI industry test scenario	S10 Climate digital-twin operations test Whole axis, Qinghe–Xiaoyue AI industry test scenario

Governance duties: separation of public service, technical operation and independent audit

Duty	Scope and coordination interface
1 Public-interest trustee	Maintains free, continuous and accessible public space with physical service entry, reviewing impacts on basic service
2 Technical operator	Runs daily scenarios, dispatch, maintenance and incident response; performance is evaluated independently
3 Independent auditor	Verifies ledger definitions, energy/carbon/water metering and privacy enforcement; issues public audit opinions
4 Residents' and users' council	Represents residents, older people, mobility-limited users, night workers and students and initiates review and improvement
5 Haidian–Ulanqab agreement working group	Negotiates task tiers, metering and incident responsibility, implemented through a signed public project agreement
6 Independent appeals and mediation body	Receives complaints, mediates disputes and supervises suspension and exit

Reversible authorization cycle and shared card rules

90-day standards and no-personal-data drill (concept)
90-day standards rehearsal: establish rules, definitions, process and ledger structure with public, synthetic or de-identified data.

One-year reversible public pilot (concept)
One-year reversible public pilot: run at selected nodes and continuously adjust facilities, rules and service from evidence.

Annual renew, modify, or stop cycle (concept)
Annual audit and renew / modify / stop decision: independent audit supports transparent evidence-led decisions.

Shared rules: every card must have a usable non-digital alternative under network, power or system failure, and every card must let a user opt out without losing basic public service. S08, S09 and S10 are the three AI industry test scenarios; until real operating data exist the ledger is published as a test specification only, with no fabricated measurements.

Fourteen synchronized spatial/design metrics (recalculated vs display value)

Metric	Machine value (metrics.json)	Display value	Meaning and limit
site_area_sqm	11412825.386 m²	approx. 11.4 km²	Working-boundary area; update when the official boundary is integrated
building_footprint_area_sqm	35851.926 m²	approx. 35,900 m²	Sum of three concept intervention envelopes, expressing the renewal action area
green_ratio	0.113434	approx. 11.3%	Union of six green concept surfaces ÷ provisional boundary area
public_space_ratio	0.044371	approx. 4.4%	Union of N1–N6 ÷ provisional boundary area
conceptual_mobility_length_m	11437.434 m	approx. 11.4 km	Sum of five concept slow-mobility links, checking structural completeness
key_area_count	3	3	Key areas
station_count	3	3	Station envelopes
wing_count	2	2	Wings
prototype_node_count	6	6	Prototype nodes
scenario_count	10	10	Operating scenarios
industry_test_count	3	3	Industry test scenarios
phase_count	3	3	Phases
declared_reversible_prototype_ratio	1.0	100% (declaration basis)	Submission/documentation coverage; built outcomes enter pilot and annual ledger
declared_manual_fallback_documentation_ratio	1.0	100% (declaration basis)	Submission/documentation coverage; built outcomes enter pilot and annual ledger

Fourteen metrics synchronize the current concept structure and display; official boundary, plan, engineering and operating inputs update them by tier.

Evidence chain: geometry → formula → metric → proposal citation

Geometry evidence	Formula	Metric
geometry/site_boundary.geojson#SITE-001	polygon_area(SITE-001) @EPSG:4548	site_area_sqm
geometry/green_space.geojson#GREEN-* ×6	union_area(GREEN-*) ÷ site_area_sqm	green_ratio
geometry/public_space.geojson#N1…N6	union_area(N1..N6) ÷ site_area_sqm	public_space_ratio
geometry/roads.geojson#ROAD-* ×5	Σ line_length(ROAD-*) @EPSG:4548	conceptual_mobility_length_m
geometry/buildings.geojson#BLDG-ST01/02/03-001	Σ polygon_area(BLDG-*)	building_footprint_area_sqm

Verifiable design targets: establish baselines, definitions and pilot data

Design target	Control quantity and formula	Value
Rest / cooling point coverage	Walking distance from any point on the spine to the nearest available point	baseline stage
Night availability of rest points	Open hours / night-period hours	baseline stage
Spine accessibility continuity	Accessible continuous length / total spine length	baseline stage
Failure recovery time	Time from degradation to recovery or completed human takeover	baseline stage
Staff arrival and help-response time	Time from call to staff arrival or answer	baseline stage
Fault repair duration	Time from fault registration to completed repair	baseline stage

Pathways for 23 professional and operating metrics

Plans and building surveys set development controls; accessibility, lighting and footfall surveys set public-service baselines; the Haidian–Ulanqab agreement and operating ledger set capacity, service and resource metrics.

The two declared_* metrics record proposal/documentation coverage for six node cards; pilot and annual ledger evidence evaluates built outcomes.